

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2353

Nickajack Dam 100 kV
August 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2353. The Interconnection Customer proposes a 297.2 MW solar plus storage facility interconnecting at the Nickajack Dam 100 kV substation in the Illinois area. The request was received on October 04, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 380 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,019,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FRANK E RATTS STATION to REDONDO 1 100kV	REDONDO 1 to N UNION CITY 100kV	104.7	136	16.4
NORTH UNION CITY to REDONDO 1 100kV	REDONDO 1 to FRANK E RATTS STA 100kV	101.1	128	18.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FRANK E RATTS STATION to REDONDO 1 100kV	Reconductor line	\$671,000	100% (\$671,000)
NORTH UNION CITY to REDONDO 1 100kV	Reconductor line	\$348,000	100% (\$348,000)
	Total allocated to project		\$1,019,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.